Francisco Cardozo

PhD, Prevention Science and Community Health
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Education

PhD, Prevention Science and Community Health. University of Miami, USA - 2025
MSc, Psychology. Universidad de los Andes, Bogotá, Colombia - 2018
BS, Psychology. Universidad Nacional, Bogotá, Colombia – 2014

Trainings

Machine Learning

1. 2025 – Artificial Intelligence and Applications in Prevention Science, SPR Conference, Seattle.
2. 2025 – Introduction of LLMs/AI Workshop. New York Data Science & AI Conference, New York.
3. 2024 – Building Generative AI Applications, PyData Conference, New York.
4. 2024 – R Software in Production, Posit::conf, Seattle.
5. 2023 – Advanced Tidymodels for Machine Learning, Posit::conf, Chicago.
6. 2023 – Advanced Package Development in R, Posit::conf, Chicago.

Statistical Methods

1. 2024 – Network Analysis using TERGM and SIENA, INSA Conference, Edinburgh.
2. 2023 – Advanced Data Analytics Certificate, Google-Coursera.
3. 2023 – Missing Data Using R, Statistical Horizons.
4. 2022 – Research Design for Causal Inference, University of Chicago.

Teaching

1. 2023 – Software Instructor, The Carpentries.
2. 2023 – Teaching Academy Certificate, University of Miami.

Honors

1. 2025, 2023 – Nominee, Graduate Teaching Assistant Award, University of Miami.
2. 2024 – Scholar Award, PyData Conference -Python.
3. 2024 – Scholar Award, Posit::conf. R software.
4. 2024 – Winner, Data Science Competition Sloboda & Bukowski Cup, SPR.
5. 2020 – Fulbright Scholarship for Doctoral Studies.
6. 2016 – Academic Scholarship, IRENEO ROSSIER, Universidad de los Andes.

Programming

R Packages

1. 2025 - DedooseR: Monitoring and Analyzing Dedoose Qualitative Data Exports.
2. 2024 – wandR: Making R learning engaging through gamified tools.
3. 2023 – BreakNBuild: Evaluate ML model performance under progressive data splits.
4. 2023 – rUM: Templates for reproducible workflows at the University of Miami.
5. 2022 – DocumentData: Tools to document datasets effectively.
6. 2022 – ChessOlympiad: Datasets for multilevel modeling practice.

Interactive web applications

1. 2024 – SNA Explorer: Visualization tool for network package dependencies and collaboration.
2. 2023 – Creating Pretty Tables in R: Interactive tutorial, University of Miami.
3. 2022 – Bias Simulation: Shiny app to illustrate reporting bias in drug studies.
4. 2022 – Reshape Data Tutorial: Long/wide transformations in R.
5. 2022 – Underage Drinking Prevention Toolkit: Public health resource site.

Skills

1. Advance level in statistical software: Mplus, Stata, Excel, SAS, SPSS.
2. Qualitative Software: Nvivo, Dedoose.
3. Advance level in programming languages: R, Python, Julia, JavaScript.
4. Reproducible Report: Quarto, Markdown, R Shiny Apps, Jupyter.

5. DevOps: Git, GitHub, Docker.
6. Web development: React, Node.js, CSS, HTML, GitHub Pages, Netlify.

Publications

Journal Articles (peer reviewed)

1. (*under review*) **Cardozo, F.,** Mehra, L., Pondikos, M., Brown, E.; Delgado, G.; Zayde, A; Gustavo, D., Rubinstein, T., Alpert, J., & Gabbay, V. (2026). Parental Early-Life and Recent Adversities as Correlates of Child Emotional and Behavioral Outcomes.
2. (in press) Morales, V., **Cardozo, F.,** Balise, R. R., St. George, S. M., & Feaster, D. J. (2026) The utility of machine learning-enhanced developmental cascade models in prevention science. Prevention Science.
3. Scoppetta, O., **Cardozo, F.,** Brown, E. C., & Morales, V. (2025). Evaluating the relationship between marijuana use, aggressive behaviors, and victimization: an epidemiological study in colombian adolescents. International journal of adolescent medicine and health, 37(1), 1–9. https://doi.org/10.1515/ijamh-2024-0167.
4. **Cardozo, F.,** Brown, E. C., Mejía-Trujillo, J., & Pérez-Gómez, A. (2023). Moderated mediation of motivation and self-efficacy in a brief motivational interviewing intervention to reduce youth alcohol use. Drugs: Education, Prevention and Policy, 0(0), 1--7. https://doi.org/10.1080/09687637.2023.2167649
5. Brown, E., Montero-Zamora, P., **Cardozo, F.,** Reyes, M., Briney, J., Mejía-Trujillo, J., & Gómez, A. (2021). A comparison of cut points for measuring risk factors for adolescent substance use and antisocial behaviors in the U.S. and Colombia. International Journal of Environmental Research and Public Health, 18, 470. https://doi.org/10.3390/ijerph18020470
6. Eisenberg, N., Brown, E., Mez, A., Mejía-Trujillo, J., Paredes-Aguilar, M., **Cardozo, F.**, Maipo, S., & Guttmannova, K. (2020). Community utilization of risk and protective factor data for prevention planning in Chile and Colombia. Health Promotion International, 2020, 1--13. https://doi.org/10.1093/heapro/daaa075
7. Reyes, M., Pinto-Gómez, C., **Cardozo, F.,** Pérez-Gómez, A., Mejía-Trujillo, J., & Toro, J. (2020). Evaluation of the prevention program brief intervention based on motivational interviewing in Colombian adolescents. International Journal of Mental Health and Addiction, 18, 1--11. https://doi.org/10.1007/s11469-019-0057-3
8. Pérez-Gómez, A., **Cardozo, F.**, & Salcedo-Tocarruncho, L. (2020). Association between psychoactive substance abuse and health disorders in Colombian problem drug users: Second study. International Journal of Mental Health and Addiction, 18. https://doi. org/10.1007/s11469-018-9881-0
9. Montero-Zamora, P., Reyes, M., **Cardozo, F.**, Brown, E., Gómez, A., Mejía-Trujillo, J., Toro, J., & Paredes, M. (2020). Adolescent substance use and its association with risk and protective factors. An exploratory analysis of the large-scale school survey of Comunidades que se Cuidan, Colombia. Adicciones, 32, 105--115. https://doi.org/10.20882/adicciones.1083.
10. Pérez-Gómez, A., Lanziano, C., Reyes, M., Mejía-Trujillo, J., & **Cardozo, F.** (2018). Perfiles asociados al consumo de alcohol en adolescentes colombianos. Acta Colombiana de Psicología, 21, 258--281. https://doi.org/10.14718/ACP.2018.21.2.12
11. Reyes, M., Mejía-Trujillo, J., Gómez, A., **Cardozo, F.**, Pinto, C., & Pinto Gómez, J. C. (2018). Effectiveness of a brief intervention based on motivational interviewing in Colombian adolescents. Psicologia: Teoria e Pesquisa, 33. https://doi.org/10.1590/ 0102.3772e33421

Books and Book chapters

1. **Cardozo, F.**, & Molano, A. (2016). Análisis estadísticos. Elementos fundamentales de análisis estadísticos en SPSS. In L. García & J. Escobar (Eds.), Análisis estadísticos. Elementos fundamentales de análisis estadísticos en SPSS (pp. 101--108). Editorial Universidad El Bosque.
2. Pérez-Gómez, A., Mejía-Trujillo, J., Reyes-Rodríguez, M. F., & **Cardozo, F.** (2015). Consumo de alcohol en menores de 18 años en Colombia: 2015. Segundo estudio en siete capitales y dos municipios pequeños. (p. 204). Dupligráficas Ltda. https://nuevosrumbos.org/website/post?id=547

Conference Presentations

1. **Cardozo, F.** (2025). Computing Inter-rater reliability for continuous scales using R. Latin American Conference on R for Research and Development. https://latinr.org/.
2. **Cardozo, F.** & Scoppetta, O. (2025). What Is Changing in Drug Use? An Analysis of Substance Use in Higher Education. Conference presentation. https://congresopsicologiacolombia.com/.
3. **Cardozo, F.** & Cañizares, C. (2025). An Introduction to Using R for Programming in Biostatistics and Data

Science, in Spanish. R/Medicine 2025 conf. https://rconsortium.github.io/RMedicine_website/.

4. **Cardozo, F.** Scoppetta, O., & Cañizares, C. (2025). From Periphery to Center? Marijuana's Changing Role in the Substance Use Network. Sunbelt 2025. International Network for Social Network Analysis. https://www.insna.org/events/sunbelt-2025--paris.
5. **Cardozo, F**. (2025). R Packages for Social Network. II Chilean Social Network Conference 2025. https://www.chisocnet.org/chisocnet-conference/conference-2025.
6. **Cardozo, F**. (2024). BreakNbuild – Tools to evaluate sample size effects on Machine Learning Models. LatinR Conference. https://latinr.org/cronograma/programa/
7. **Cardozo, F.,** Eric C. Brown., Scoppetta, O. (2024). Evaluating the Impact of Marijuana on Adolescent Aggression: A Population-Based Study in Colombia. Fifth Annual Graduate and Postdoctoral Research Symposium. https://events.miami.edu/event/research_symposium_2024
8. **Cardozo, F.** (2023). Exploring the interrelationships between NIH-funded projects on adolescent alcohol use: A two-mode network analysis. Sunbelt 2023. International Network for Social Network Analysis. https://www.insna.org/events/ sunbelt-2023
9. **Cardozo, F.** (2023). Exploring the similarities in parent-child perceptions of drug use risk: The role of genetics and environment. Fifth Annual Graduate and Postdoctoral Research Symposium. https://events.miami.edu/event/ research_symposium_2023
10. Pinilla, V., **Cardozo, F.,** Montero-Zamora, P., Brown, E.C. (2022). Businesses that Care: A Toolkit for Implementing Underage Drinking Prevention Programs in Communities. Oral presentation at the Social Marketing Conference, Clearwater Beach, FL.
11. Balise, R., Nahodyl, L., Calderon, A., & **Cardozo, F.** (2022). How to create publication ready tables in R. Presented at UseR conference. https://user2022.r-project.org/program/tutorials/
12. Brown, E.C., Oliveira Corrêa, A., **Cardozo, F**., Guidi-Gentil, C., Montero-Zamora, P., Wells, A. (2022). A Systematic Review of Preventive Interventions for Youth Alcohol, Tobacco, and Other Drug Use in Latin America and the Caribbean. Oral presentation at the Annual Meeting of the Brasilian Society for Prevention Research, online.
13. **Cardozo, F.**, & Molano, A. (2021). Ecological factors in friendship relationships and alcohol use in scholars. Presented at Networks 2021: A Joint Conference of Sunbelt and NetSci. https://doi.org/10.13140/RG.2.2.17630.28487

Posters (presenter)

1. **Cardozo, F.,** Ivanova, Miglena., Cañizares, Catalina, Arrington, Myles. (2025) Mapping Adolescent Socio-Emotional Well-Being: A Network Analysis of Future of Families and Child Well-being Study. SRCD 2025 Biennial Meeting.
2. **Cardozo, F.,** & Brown, C. (2024). Data Science and Machine Learning Methodology Dissemination in Latin America. Frost Institute for Data Science and Computing.
3. **Cardozo, F.** (2024). Polydrug Use Patterns Among Colombian Adolescents: A Graph-Based Exploration. International Network for Social Network Analysis. https://sunbelt2024.com
4. **Cardozo, F.,** & Brown, C. (2023). Data Science and Machine Learning Methodology Dissemination in Latin America. Frost Institute for Data Science and Computing.
5. **Cardozo, F.**, Brown, C., Mejía-Trujillo, J., & Pérez-Gómez, A. (2023). Parent-Youth Correspondence in Perceptions of Risk Factors for Adolescent Drug Use: A Dyadic Analysis. Poster presented at Society for Prevention Research Annual Meeting, SPR. DOI: 10.13140/RG.2.2.25548.59521
6. **Cardozo, F.**, Montero-Zamora, P., Brown, C., Ortiz-García, J., & Beaulieu, D. (2023) Cost-Effectiveness Analysis of the Tomando Buenas Decisiones Program to Reduce Family Risk Factors Associated with Alcohol Use. Poster presented at Society for Prevention Research Annual Meeting, SPR. DOI: 10.13140/RG.2.2.28904.03848
7. **Cardozo, F.**, Brown, E., Mejía-Trujillo, J., & Pérez-Gómez, A. (2021). Moderated mediation of motivation and self-efficacy in a brief motivational interviewing intervention to reduce youth alcohol use. Poster presented at Society for Prevention Research Annual Meeting, SPR. https://doi.org/10.13140/RG.2.2.17630.28487
8. **Cardozo, F.**, & Molano, A. (2019). Close friends vs. The norm effect: Effects of proximal and distal alcohol social context. Poster presented at Society for Prevention Research Annual Meeting, SPR. https://doi.org/10.13140/RG.2.2.11232.23048
9. **Cardozo, F.**, Guerrero, P., & Molano, A. (2017). Time-variant effects of friendship change on academic achievement: An exploratory social network analysis in Colombia. Poster presented at Society for Research in Child Development, SRCD. https://doi.org/10. 13140/RG.2.2.17630.28487

Posters and presentations (coauthor)

1. Serban, F., **Cardozo, F**., & Brown, E. C. (2026). Application of Generative Artificial Intelligence in Teaching Prevention Science. Seven Annual Graduate and Postdoctoral Research Symposium. https://www.grad.miami.edu/about/grad-events-and-calendar/research-symposium/index.html.
2. Serban, F., **Cardozo, F.**, & Brown, E. C. (2025). Development of a generative artificial intelligence–based approach to teaching prevention science. European Society for Prevention Research Annual Meeting, Berlin, Germany.
3. Guidi, C., **Cardozo, F**., Thurow, C., Ribeiro-Schneider, D., Regina-Noto, A., & Brown, E. (2025). Measurement Invariance by Sex in Family Risk Factors for Adolescent Alcohol and Drug Use in Brazil. Poster session presented at the annual meeting of the Society for Prevention Research, Seattle, W.
4. Pondikos, M., **Cardozo, F.**, Mehra, L., Delgado, G., Brown, E., Zayde, A., Derella, O., Rubinstein, T., Alpert, J., Gabbay, V., (2025). Parental Early-Life and Recent Adversities as Correlates of Child Emotional and Behavioral Outcomes. Presented at PRIME Research Day, Montefiore Medical Center / Albert Einstein College of Medicine.
5. Delgado, G., Gallo, L., Bhambhani, Y., **Cardozo, F.**, Holzman, S., Musicaro Yuksel, R., Hribar, B., Perez, H., & Gabbay, V. (2025). Predictors of engagement in a randomized controlled trial investigating opioid use disorder and chronic pain treatment. Presented at PRIME Research Day, Montefiore Medical Center / Albert Einstein College of Medicine.
6. Guidi, C., **Cardozo, F.**, & Brown, E. (2024). Analysis of the Brazilian Mortality National Data System (SIM) for Youth Suicide Prevention. Poster session presented at the annual meeting of the Society for Prevention Research, Washington, D.C.
7. Guidi, C., **Cardozo, F**., & Brown, E. (2024). Risk Factors for Youth Substance Use in Brazil: A Multigroup Confirmatory Factor Analysis Comparing Age and Gender Groups. Poster session presented at the annual meeting of the Society for Prevention Research, Washington, D.C.
8. Guidi, C., **Cardozo, F.,** Ribeiro-Schneider, D., & Brown, E. (2022). Communities That Care – Brazil: Collection of Youth Risk and Protective Factor Data in Florianopolis. Poster presented at Society for Prevention Research, , Washington, D.C.
9. Montero-Zamora, P., Reyes-Rodríguez, M.F., **Cardozo, F.,** Brown, E.C., Pérez-Gómez, A., Mejía-Trujillo, J., Toro, J., Paredes-Aguilar, M. (2018). Concurrent validity of adolescent risk and protective factors for substance use in Colombia. Poster presentation at the Annual Meeting of the Society for Prevention Research, Washington, DC. Honorable mention.

Teaching

Instructor

1. 2026 spring. EPH 752: Advanced Research Methods. Public Health. University of Miami
2. 2015 – 2018. PSI 122: Quantitative Methods III. Department of Psychology, Universidad de El Bosque. Colombia.
3. 2015 - 2018 PSY 131: Program evaluation. Department of Psychology, Universidad de El Bosque. Colombia.

Teaching Assistantships

1. 2024 Spring. EPH 752: Advanced Research Methods. Department of Public Health Sciences, University of Miami.
2. 2023 June. R Medicine 101: Introduction to Clinical Data Analysis with R. R-consortium.
3. 2023 Summer. BST 692: Data Science and Machine Learning for Health Research. Department of Public Health Sciences, University of Miami.
4. 2023 Spring. BST 623: Software Tools for Manuscript Development and Reproducible Research. Department of Public Health Sciences, University of Miami.
5. 2022 Spring EPH 752: Advanced Research Methods. Department of Public Health Sciences, University of Miami.
6. 2022 Summer BST 692: Data Science and Machine Learning for Health Research. Department of Public Health Sciences, University of Miami.
7. 2021 Fall BST 623: Software Tools for Manuscript Development and Reproducible Research. Department of Public Health Sciences, University of Miami.
8. 2021 Spring. EPH 652: Research Methods. Department of Public Health Sciences, University of Miami.
9. 2016 – 2018. EDUC 4026: Advanced Research Methods. Department of Education, Universidad de Los Andes. Colombia.

Guest Lecturer

1. 2025 CISO 4604: Program Evaluation: Causal Inference. Universidad de Los Andes. Colombia. Colombia.
2. 2024 EPH 752: Advanced Research Methods: Software for advanced Methods. University of Miami. USA.
3. 2024 CISO 4604: Program Evaluation Causal Inference. Universidad de Los Andes. Colombia. Colombia.
4. 2023 CBPC 1069: Education in Colombia: What is prevention science. University of Los Andes. Colombia.
5. 2023 CBPC 1066: Social Policy Experiments: Causal Inference in Public Health. University of Los Andes. Colombia.
6. 2023 CISO 4604: Project Evaluation: Causal Inference. Universidad de Los Andes. Colombia. Colombia.
7. 2022 EPH 752: Advanced Research Methods: Software Integration for Data Analysis. University of Miami. USA.
8. 2022 EPH 752: Advanced Research Methods: Dyadic Data Analysis. University of Miami. USA.
9. 2022 EPH 752: Advanced Research Methods: Monte Carlo Simulation to Estimate Power. University of Miami.
10. 2022 EPH 701: Research Methods: Causal Inference in Prevention Science. University of Miami. USA.
11. 2019 EDUC 4026: Advanced Quantitative Methods: Moderation. University of Los Andes. Colombia.
12. 2017 EPH 752: Advanced Research Methods: Social Network Analysis. University of Miami. USA.
13. 2015 EPH 752: Advanced Research Methods: Confirmatory Factor Analysis. University of Miami. USA.

Invited Talk

1. 2025-03-12 Fostering Open Science in Latin America- rOpenSci.

Workshops Designed and Delivered

1. 2026-01 Machine Learning for Social Sciences. Universidad de los Andes.
2. 2025-11 Introduction to the Tidymodels ecosystem for Machine Learning. Latin American Conference on R for Research and Development. Preconference Workshop.
3. 2024-05 Programming with Python. Universidad de Los Andes. Colombia. Funded by: University of Miami. Software Carpentry.
4. 2024-03 R for Reproducible Research. University of Miami. USA. Organized by: University of Miami. Software Carpentry.
5. 2024-02 Machine Learning for Classification Problems. Fundación Universitaria Konrad Lorenz. Colombia. Funded by: Frost Institute for Data Science and Computing.
6. 2024-02 Programming with Python. University of Miami. USA. Funded by: University of Miami - Software Carpentry.
7. 2024-02 Machine Learning Modeling with the R Tidyverse Ecosystem. Florida International University. USA. Organized by: Florida International University.
8. 2024-01 Mastering Effective Data Visualization with ggplot2 in R: A Hands-on Workshop for Social Work Researchers. USA. Presented in The Society for Social Work and Research Conference.
9. 2023-12 Imbalanced Data in Machine Learning Modeling. University of Miami. USA. Funded by: Frost Institute for Data Science and Computing.
10. 2023-11 Introduction to Machine Learning. International Seminar for the Evaluation of the Quality of Education (ICFES). Colombia. Funded by: Frost Institute for Data Science and Computing.
11. 2023-11 Creating Tables in R. Universidad Nacional. Colombia. Funded by: Frost Institute for Data Science and Computing.
12. 2023-05 Regression Techniques for Classification Problems. Universidad de los Andes. Colombia. Funded by: Frost Institute for Data Science and Computing.

Memberships

1. 2026 Royal Statistical Society.
2. 2025 R-Medicine. Member of the director's board.
3. 2025 Society for Research in Child Development SRCD.
4. 2023 rOpenSci. Transforming science through open data, software & reproducibility.
5. 2023 Software Carpentry. Teaching researchers the computing skills.
6. 2022 Society for Prevention Research SPR.
7. 2021 International Network for Social Network Analysis INSNA.

Professional Service

1. 2025 Reviewer, Prevention Science.
2. 2025 Reviewer, Journal of Behavioral Medicine.

Research Projects

Current

Development of an LLM-Based chatbot for Program Implementation Fidelity

I led the development of an AI-powered system to support implementation fidelity for the IBEM (Brief Intervention in Motivational Interviewing) program. This work involved designing and evaluating a Large Language Model based chatbot to simulate realistic participant interactions, as well as creating an automated feedback platform that provides facilitators with tailored, actionable insights on their adherence to the program steps. The system draws on data from facilitator–chatbot interactions to generate fidelity reports, refined through mixed-methods research including expert feedback, certified implementer input, and practitioner surveys and focus groups. By integrating motivational interviewing principles with scalable AI technologies, this project demonstrates how automated tools can enhance training, improve fidelity, and expand the reach of evidence-based preventive interventions.

Role: Principal Investigator

Evaluating a Multimodal Parenting Intervention to Address COVID-19 Impact in the Bronx

This NIH-funded project evaluates the psychological and healthcare access impact of a parenting intervention for vulnerable caregivers in the Bronx following the COVID-19 pandemic. The study uses a randomized controlled trial to assess the effects of a 12-session Mentalization-Based Therapy (CARE) program and a HIPAA-compliant mobile health app (Valera) on stress, healthcare access, and reflective functioning among caregivers of children with psychiatric or autoimmune conditions, and frontline healthcare workers. As the project's Data Scientist, I developed and maintained workflows for data extraction from REDCap via API, implemented reproducible pipelines for instrument scoring and validation, and led the preparation of analysis-ready datasets. I also collaborated on study design proposals, advised on statistical methodology, and contributed to manuscript development. My role emphasizes the integration of data science into the research cycle, enabling efficient data use, reproducible computation, and high-impact insights for improving mental health in underserved populations.

Funding Source: NIH PAR-20-237

Role: Postdoctoral Fellow - Data Scientist

Past

2023 Expanding the Use of Collaborative Data Science at UM: Machine Learning Methodology Dissemination in Latin America

An initiative funded by the Institute for Data Science and Computing; this project aimed to strengthen collaborative data science networks across Latin America. Through targeted workshops on machine learning training and resource dissemination, I outreach to multiple institutions to foster data-driven research and capacity-building throughout the region.

Funding Source: Institute for Data Science and Computing.

Grant: # CD230007

Amount: $ 20,000.00 USD.

Role: Co-PI

2021 Organization of American States (OAS) - Inter-American Drug Abuse Control Commission (CICAD) and the University of Miami (UM): First Hemispheric Drug Use Prevention Forum

Contributed to the design, collection, and organization of a comprehensive dataset on drug use prevention interventions across Latin America and the Caribbean, supporting regional efforts to understand and address substance use challenges.

Funding Source: Copeland Foundation Award (DPHS Internal Application).

Grant #GR0153355 (Award # CFA02-2020).

Role: Graduate Research Assistant.

Languages

Spanish, English